

Time Series Foundation Models: An Introduction

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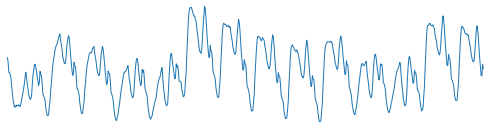
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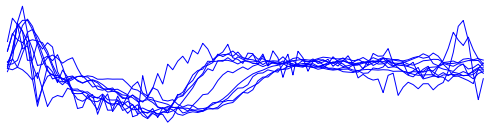
1. Introduction
2. The First Generation of TSFMs
3. The Second Generation of TSFMs
4. Conclusion & Hands-on session

Introduction

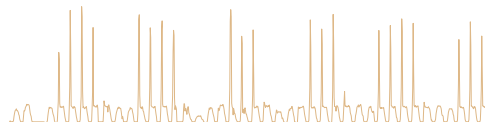
Standard paradigm: one model per dataset



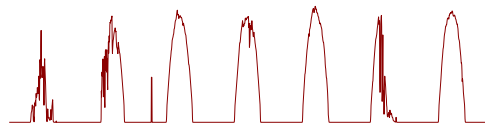
Learn f_{θ_1} on aggregated load curves



Learn f_{θ_3} on heatbeat signals

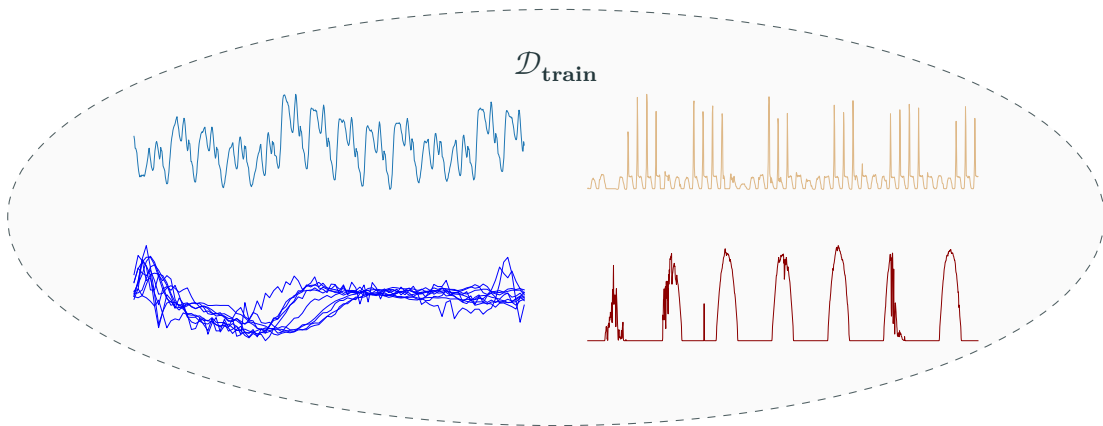


Learn f_{θ_2} on road occupancy rates

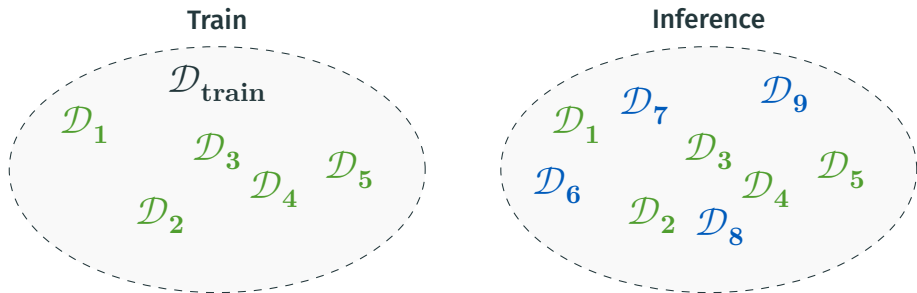


Learn f_{θ_4} on PV power production

A shift with Foundation Models: joint learning



About generalization at inference



Two types of generalization at inference

- (i) **In-Domain (ID):** new series sampled from the pretraining corpus ($\mathcal{D} \subset \mathcal{D}_{train}$)
- (ii) **Out-of-Domain (OOD):** entirely new datasets, not seen during training ($\mathcal{D} \not\subset \mathcal{D}_{train}$) (0-shot)

What can we hope from Foundation Models?

A model able to (i) **memorize patterns** + (ii) to **adapt** to new time series

What can we hope from Foundation Models?

A model able to (i) **memorize patterns** + (ii) to **adapt** to new time series

→ A single model that can be deployed on multiple domains

→ Strong performances on new datasets

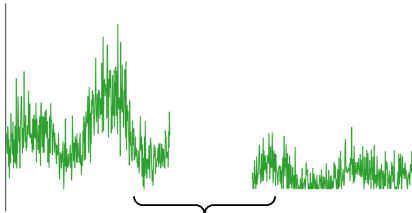
→ Emerging abilities beyond memorization

What impact in practice?

HIGH-VALUE SCENARIO #1

Lack of data (cost, cold start etc.) to train a good model

Aucun
historique



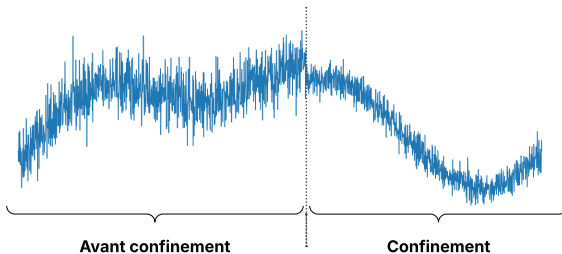
Bloc manquant

Cold-start scenario: installing a new sensor.

What impact in practice?

HIGH-VALUE SCENARIO #2

Tackling **distribution shifts** in the observed data



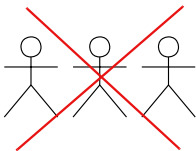
Distribution shift: impact of a lockdown on load demand.

What impact in practice?

HIGH-VALUE SCENARIO #3

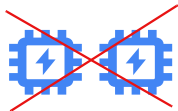
Data is available, but (i) a complex problem + (ii) **missing hardware resources** or (iii) **no expertise in machine learning**

→ Need for a model **ready to use**, running **fast**, applicable to a wide range of TS



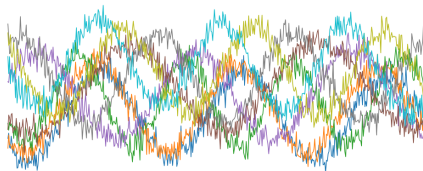
Pas de modélisateurs

Ou



Peu de moyens de calcul

Mais



De nombreux besoins

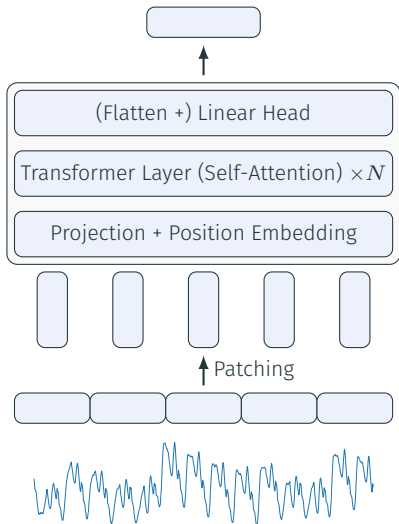
What impact in practice?

One-sentence summary

Foundation Models bring a **universal base** for **forecasting** or **imputation**: they compensate for **lack of data**, are robust to **distribution shifts** and democratize the **use of advanced methods**

The First Generation of TSFMs

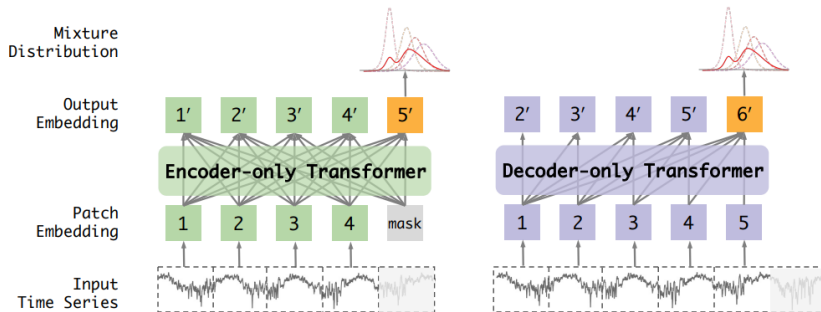
PATCHTST (2023), self-attention meets time series forecasting



- Until 2023: Transformers achieve great successes in sequence modelling... but not in TS modelling
 - E.g. *Are Transformers effective for TS forecasting?* [14]
- **PATCHTST** introduced the first efficient design for Transformer-based TS forecasting [8]
- **Key idea #1:** segmenting a TS into **patches** → tokens to the Transformer
- **Key idea #2: channel-independence**
- This design still powers most of modern TS architectures

The first wave of TSFMs are mostly patch-based Transformers

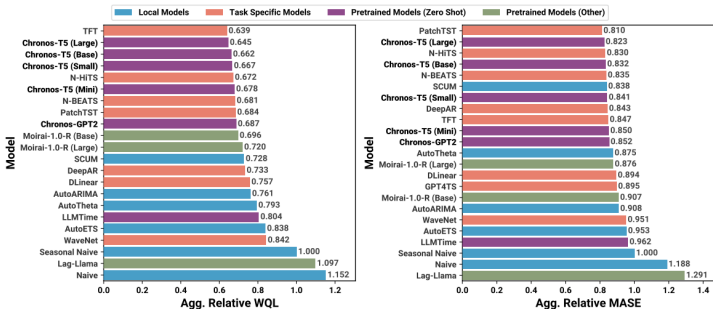
TIMESFM [4], MOIRAI [12], ...: patch-based Transformers learned jointly on multiple datasets



Architectures of baseline TSFMs [13].

The first wave of TSFMs are mostly patch-based Transformers

TIMESFM [4], MOIRAI [12], ...: patch-based Transformers learned jointly on multiple datasets



Typical results: CHRONOS zero-shot benchmark [3].

Limitations

FIRST GENERATION OF TSFMS FOR FORECASTING MAIN LIMITATIONS

1. Lack of rigorous unified benchmarks [6] - GITEVAL (Dec. 2024) as a 1st effort [1]
2. Struggle to outperform the best supervised baselines / may require finetuning [5]
3. **Univariate-only** models, unable to use **additional information (covariates)**

The Second Generation of TSFMs

- How to **integrate covariates** (if available) at inference ?

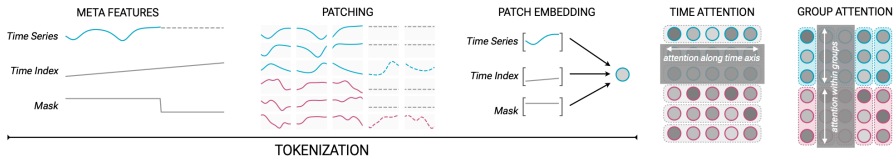
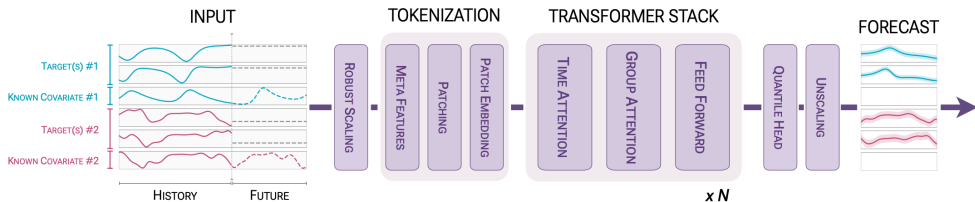
THE SECOND GENERATION OF TSFMS

Chronos2 [2] (Amazon)

TiRex2 [9] (NXAI)

TS-ICL (EDF R&D) [7]

CHRONOS2 mechanism [2]



CHRONOS2 architecture. Plot from [2].

CHRONOS2 Insights

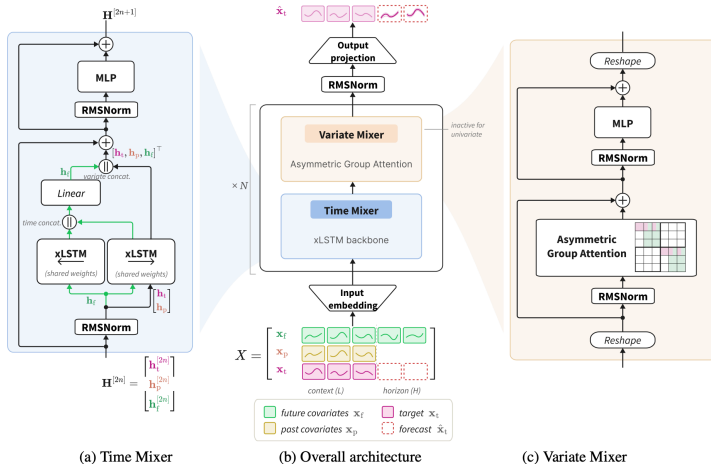
How does CHRONOS2 handle covariates?

1. Uses an **In-Context Learning** + Group Attention mechanism to incorporate covariates
2. Relies on a **synthetic covariates prior during training** (implementation not publicly available)
3. Strongly **inspired by Tabular Foundation Models**

CHRONOS2 Strengths

1. **State-of-the-art performance** across TSFM forecasting benchmarks (GiftEval [1], FEVBench [11], Time [10])
2. **Very fast inference**: around **1 second** for **100 windows** (thousands of points) on an **H100 GPU**
3. **Efficient integration of covariates**, enabling improved predictive performance

TiREX 2 mechanism [9]



TiREX2 architecture. Plot from [9].

TiREX 2 Insights

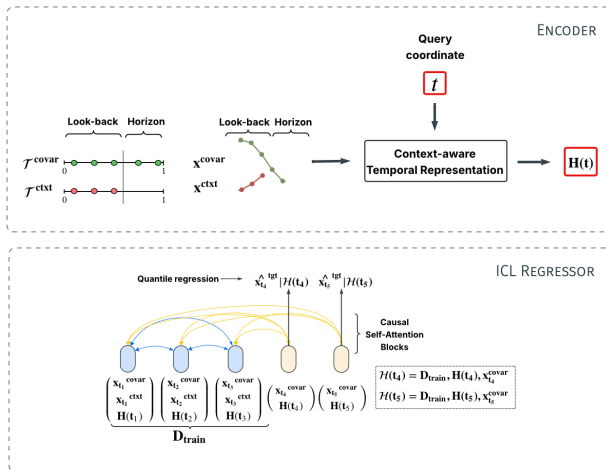
How does TiREX2 handle covariates?

1. **Recurrent xLSTM backbone**: alternates a **Time Mixer** (xLSTM) and a **Variate Mixer** (group attention)
2. **Future-known covariates read bidirectionally**, while targets and past covariates stay **strictly causal**
3. **Asymmetric attention mask**: targets attend to covariates, never the reverse, so there is no future leakage
4. **Documented synthetic coupling prior** (linear/non-linear SCM, cointegration, functional links)

TiREX2 Strengths

1. **State-of-the-art zero-shot** on FEVBench [11] and GiftEval [1], ahead of CHRONOS2
2. **Streaming inference**: constant cost per new patch, **linear** in context length (vs. quadratic for CHRONOS2)
3. **Compact**: 38M parameters (univariate) + 44M (multivariate mode)

TS-ICL mechanism [7]



TS-ICL architecture

TS-ICL Insights

How does TS-ICL handle covariates?

1. **Time-indexed, no patching:** (timestamp, value) pairs compressed into 32 latent tokens (Perceiver encoder)
2. **Pointwise in-context regression:** observed points act as training set, query timestamps as test set
3. **Causal DAG/SCM prior** producing informative, redundant or independent covariates

TS-ICL Strengths

1. **Forecasting + imputation** in one model; **SOTA zero-shot imputation**, with and without covariates
2. **Robust to missing / irregular data:** at 50% missing look-back, MASE 1.89 vs. 2.44 for CHRONOS2
3. **Light:** 27M parameters, $\sim 40\times$ faster than tabular FMs (TabPFN-TS)
4. *Trade-off:* $\sim 4\times$ slower than CHRONOS2, which remains best on covariate forecasting

Conclusion & Hands-on session

Conclusion

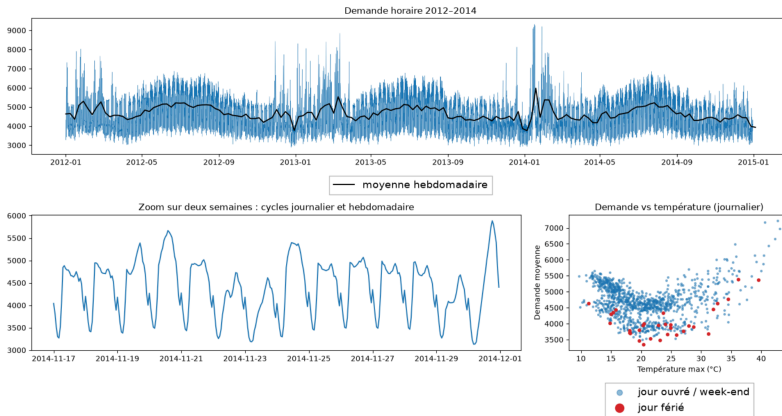
Where do TSFMs stand today?

1. **Strong zero-shot performance:** TSFMs now often **outperform supervised models**, especially in **low-data regimes**
2. **Fine-tuning** remains possible to **further improve** performance on a given use case
3. **Beyond forecasting:** promising results on **imputation**, **classification**, covariate-aware tasks, etc.

Take-home message

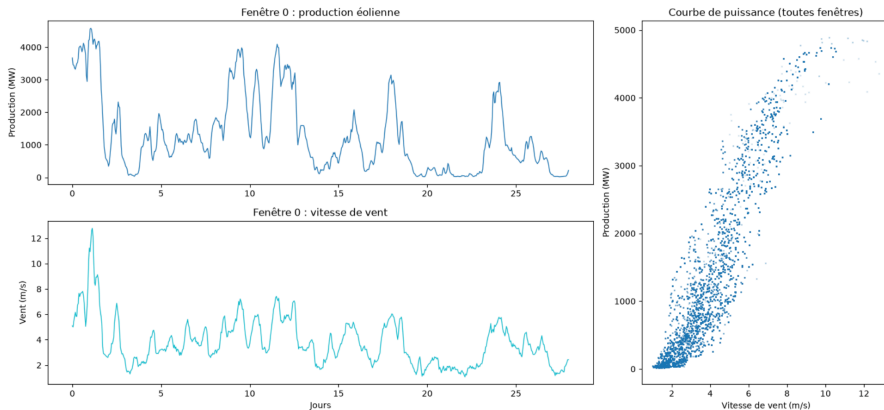
1. **Very active field:** new models and benchmarks released **every week**
2. TSFMs are **tools we need to master** and **integrate into our pipelines**

Hands-on session: Forecasting



Day-ahead forecasting of electricity consumption in Victoria (Australia)

Hands-on session: Imputation



Imputation of missing days in wind power generation curves

Hands-on session

<https://github.com/EtienneLnr/workshop-enedis-tsfm/tree/main>

THANKS FOR LISTENING!

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